EVB BGS12P2L6 User Guide

Evaluation Board BGS12P2L6

About this document

Scope and purpose

This measurement report provides an overview of the evaluation board (EVB) of the BGS12P2L6 SPDT Switch. In this document, the BGS12P2L6 circuit schematic, PCB layout and measurement results are summarized.

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1 BGS12P2L6 Description

The BGS12P2L6 is a general-purpose RF MOS power switch, designed to cover a broad range of high power applications from 0.05 to 6 GHz, mainly in the transmit path of GSM, WCDMA, and LTE mobile phones. The chip integrates on-chip CMOS logic driven by a simple, single-pin CMOS or TTL compatible control input signal. Unlike GaAs technology, external DC blocking capacitors at the RF ports are only required if DC voltage is applied externally. The BGS12P2L6 RF switch is manufactured in Infineon's patented MOS technology, offering the performance of GaAs with the economy and integration of conventional CMOS including the inherent higher ESD robustness. The device has a very small size of only 0.7 x 1.1 mm 2 and a maximum height of 0.31 mm.
[In order to get the EVB BGS12P2L6, place your order in either the Integrated Distribution Solution (IDIS) tool, Infineon Sample Request (ISaR) tool or contact your business partner using the reference number: SP002043106].

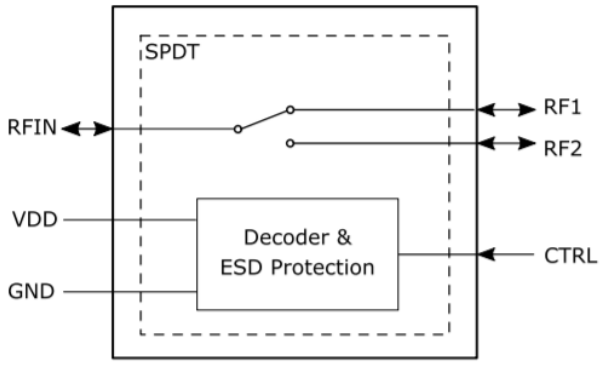


Figure 1 BGS12PL6 Block Diagram

1.1 GPIO Specification Block Diagram

The device is operated by General Purpose Input and Output (GPIO) control (use the right V_{CTRL} based on the datasheet). Table 1 shows the modes of operation.

		Control Inputs
State	**Mode**	**CTRL**
1	RFIN-RF1	0
2	RFIN-RF2	1

Table 1 Modes of operation (Switch Control Truth Table)

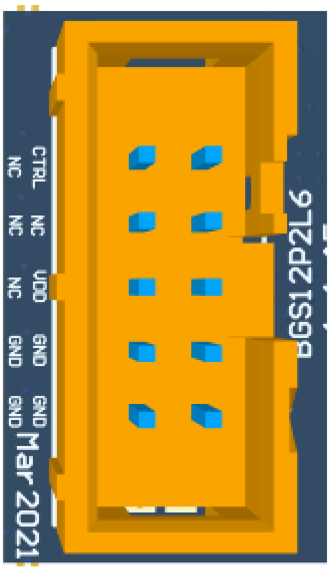
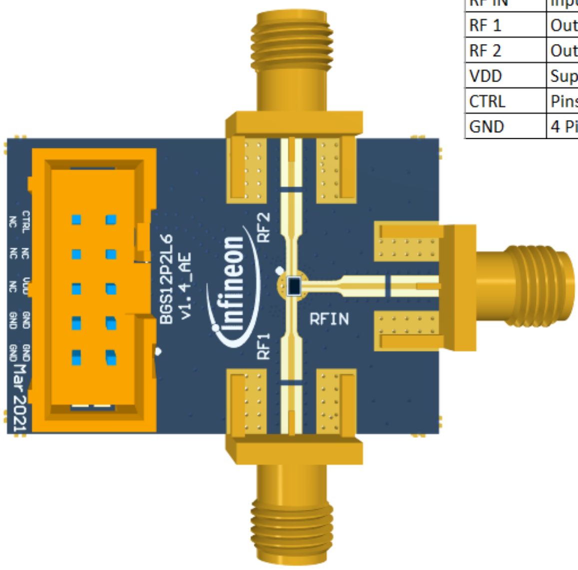


Figure 2 GPIO Connector with pin designations

2 Evaluation Board

The evaluation board was designed to ease customer evaluation of the BGS12P2L6. All transmission lines and SMA connectors have a load of 50Ω. The 10-pin connector is used to power and control the BGS12P2L6. The RF lines are isolated from the GPIO control lines.



PIN	DESCRIPTION
RF IN	Input of RF Signal
RF 1	Output 1
RF 2	Output 2
VDD	Supply Voltage, 1.8V
CTRL	Pins to control the SPDT, 1.8V
GND	4 Pins used as ground

Figure 3 Evaluation Board

2.1 PCB Layout

The total thickness of the board is ~1mm and its layers' composition is presented in the image below.

Layer	Material	Via / DK-Bohr.	Dicke
LS-Lack			20
T	Cu-Folie		35
	1 x 106 (Prepreg - Panasonic R-5670 Typ 3313)		106
L2			35
	1 x 100 (Kern - Panasonic Megtron6 R-5775)		100
L3			35
	1 x 106 (Prepreg - Panasonic R-5670 Typ 3313)		106
	1 x 100 (Kern - Panasonic Megtron6 R-5775)		200
	1 x 106 (Prepreg - Panasonic R-5670 Typ 3313)		106
	1 x 100 (Kern - Panasonic Megtron6 R-5775)		100
	1 x 106 (Prepreg - Panasonic R-5670 Typ 3313)		106
B	Cu-Folie		35
LS-Lack			20
		Gesamtdicke:	1004 µm

Figure 4 PCB Layout

2.2 Schematic

Figure 4 shows the schematic of the Evaluation Board.

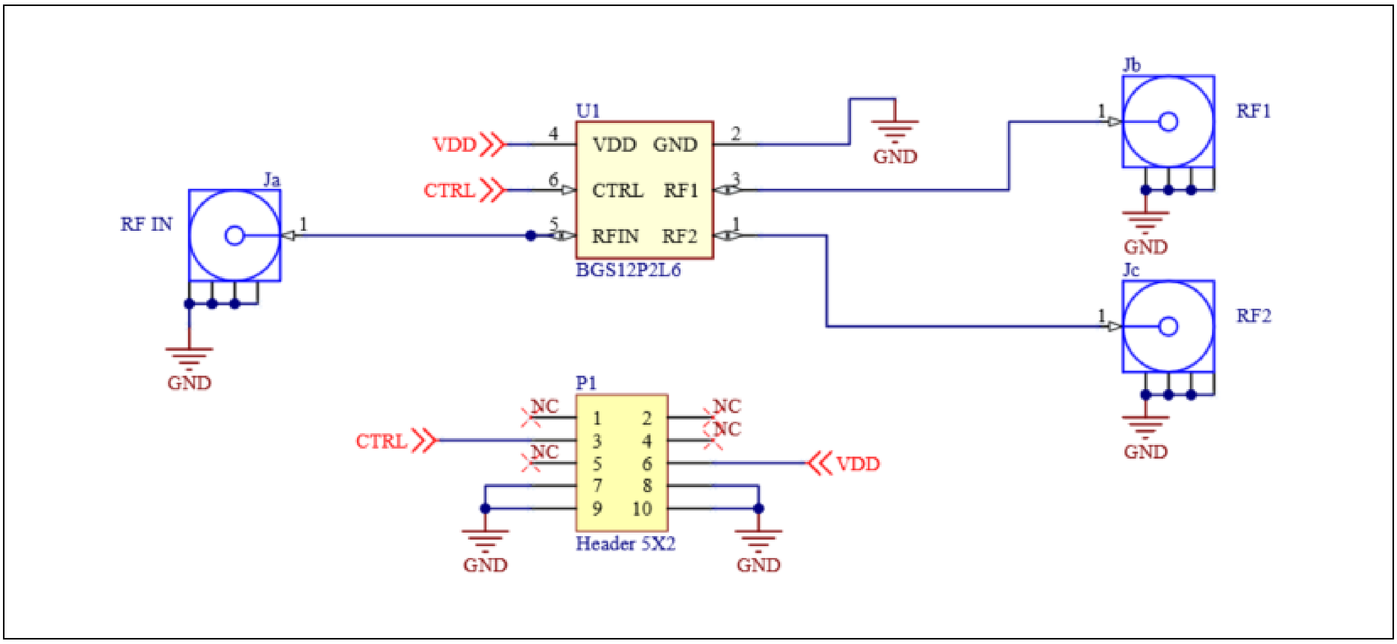


Figure 5 BGS12P2L6 EVB Schematic

2.3 Bill of Materials

Designator	Value	Description
RF1,RF2,RFIN	Jc, Jb, Ja	SMA Connectors
Connector	P1	10 Pin Connector, Header 5x2
U1		BGS12P2L6 SPDT Switch
PCB		BGS12P2L6 Evaluation Board

2.4 Evaluation Board Hardware

The BGS12P2L6 kit contains a BGS12P2L6 evaluation board and a through transmission line. The board allows the user to connect RF signals to the device via SMA connectors. The user controls the BGS12P2L6 using the CTRL pin and setting the proper voltage. In order to subtract the losses of PCB traces and have an accurate measurement, de-embedding must be done first. The de-embedding file can be found on Infineon's website (See link at the bottom of this document).

2.5 RF Connectors

It is recommended to have all SMA Ports connected to the Network Analyzer (VNA) during measurements. If there are not enough ports available on the VNA, the remaining SMA connectors of the device must be terminated with a 50Ω load (see image below).

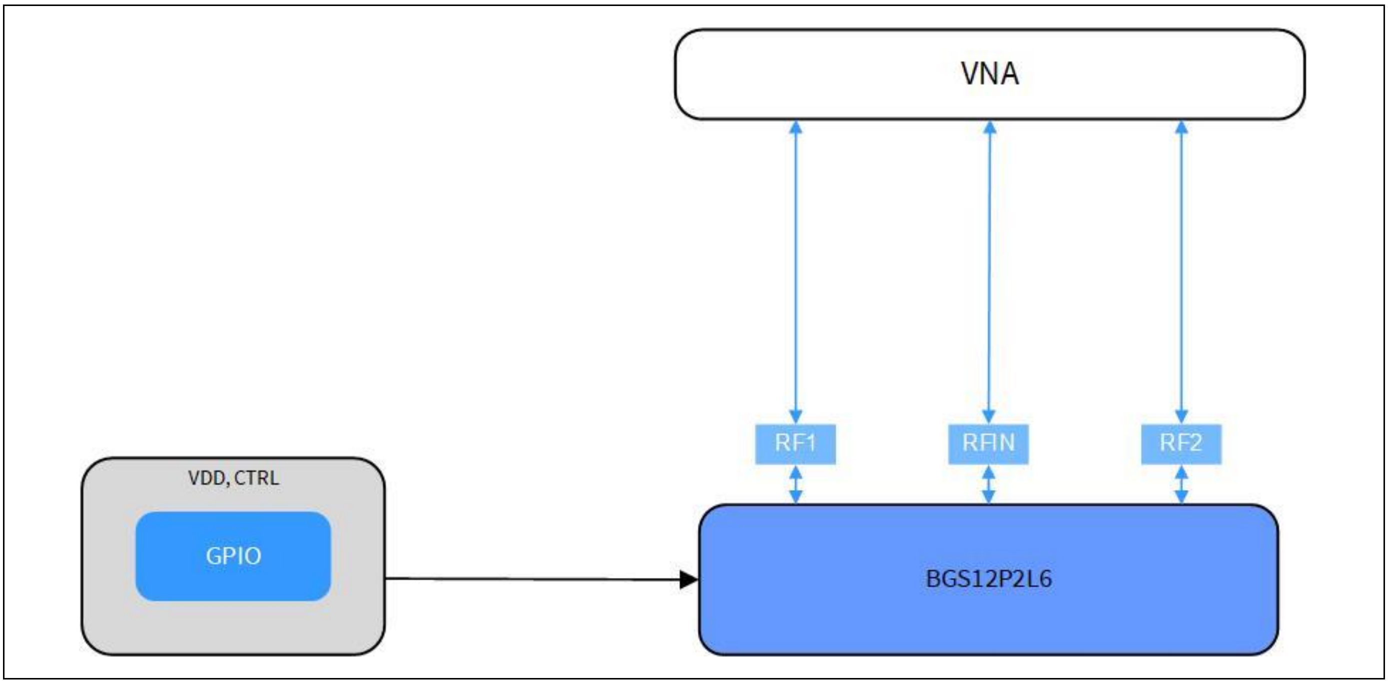


Figure 6 Measurement Setup Schematic

3 S-Parameters

Performance plot for frequencies between up to 6 GHz are shown in the image below. All RF measurements include de-embedding. The S-parameter files including the de-embedding can be found on Infineon's website (See link at the bottom of this document).

RF Characteristics at TA = 25 ◦C, $Power_{RF}$ = 0 dBm, V_{DD} = 1.8 V

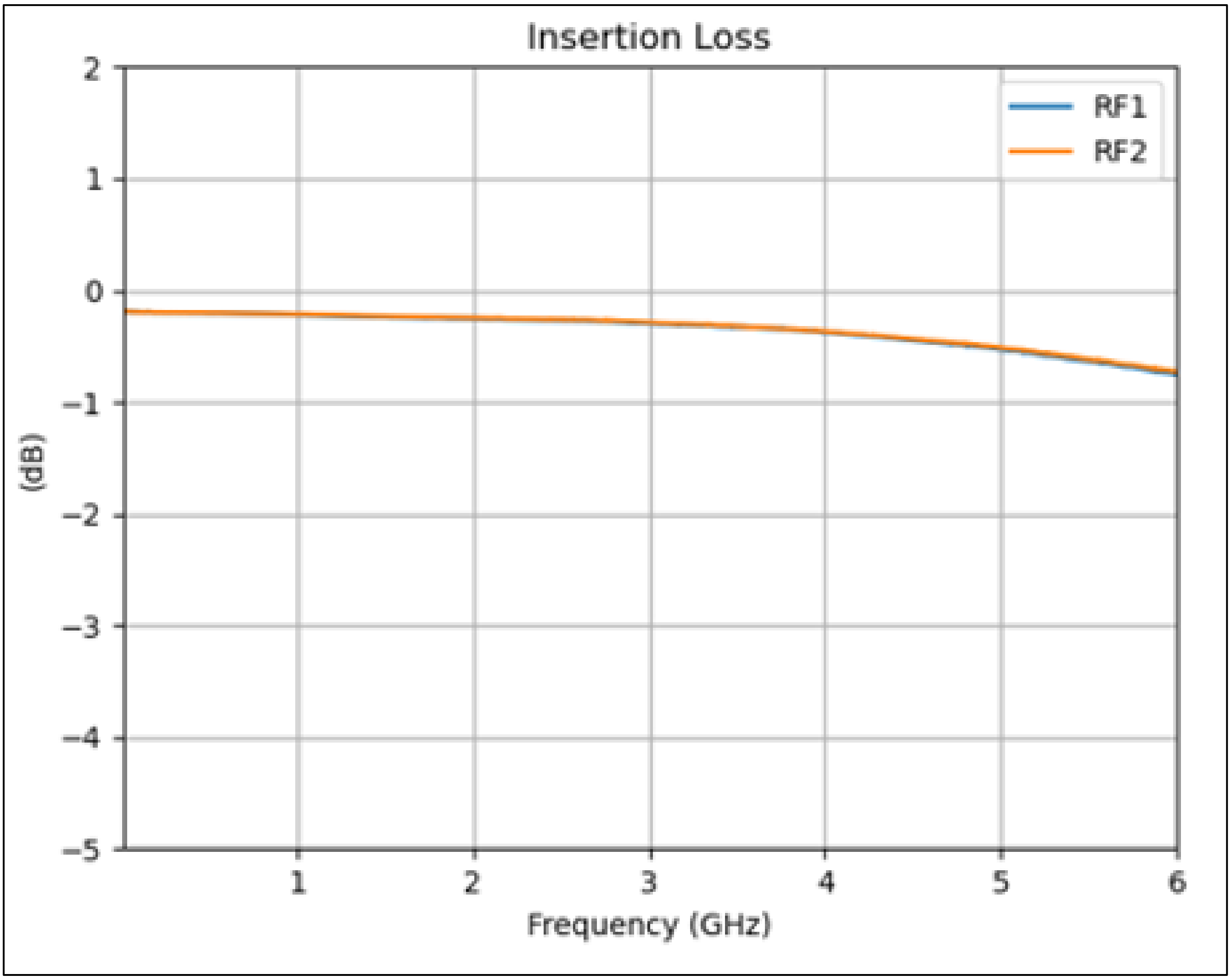


Figure 7 Insertion Loss

	Insertion Loss (dB)						
Frequency (GHz)	**0,96**	**2,17**	**2,7**	**3,8**	**4,2**	**5**	**5,925**
RF Path							
RF1	0,2	0,25	0,28	0,34	0,37	0,44	0,56
RF2	0,19	0,24	0,27	0,33	0,35	0,41	0,52

Table 2 Insertion Loss

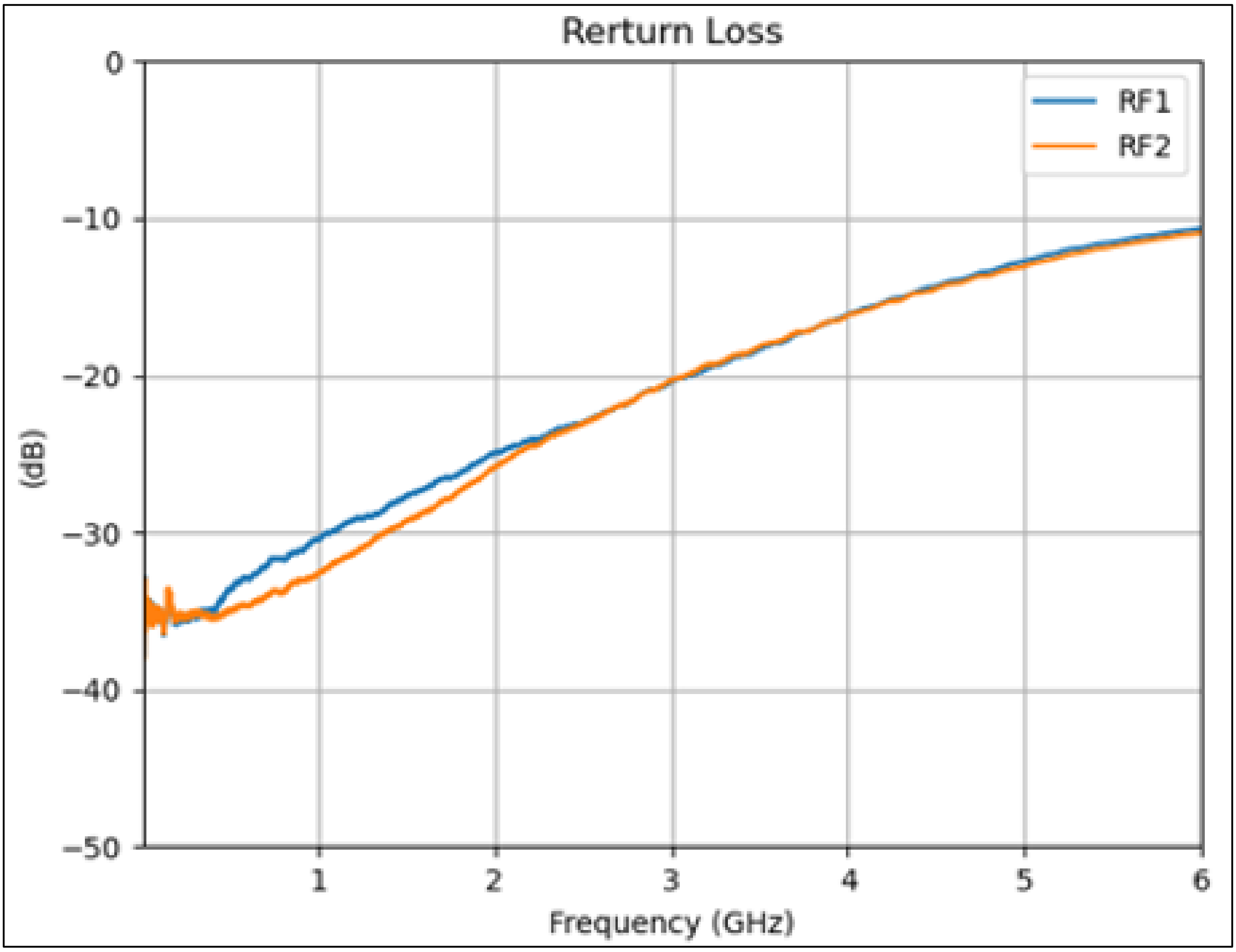


Figure 8 Return Loss

	Return Loss (dB)						
Frequency (GHz)	**0,96**	**2,17**	**2,7**	**3,8**	**4,2**	**5**	**5,925**
RF Path							
RF1	27,68	21,4	19,53	17,17	16,43	15,22	13,8
RF2	27,83	21,89	20,08	17,78	17,2	16,03	14,65

Table 3 Return Loss

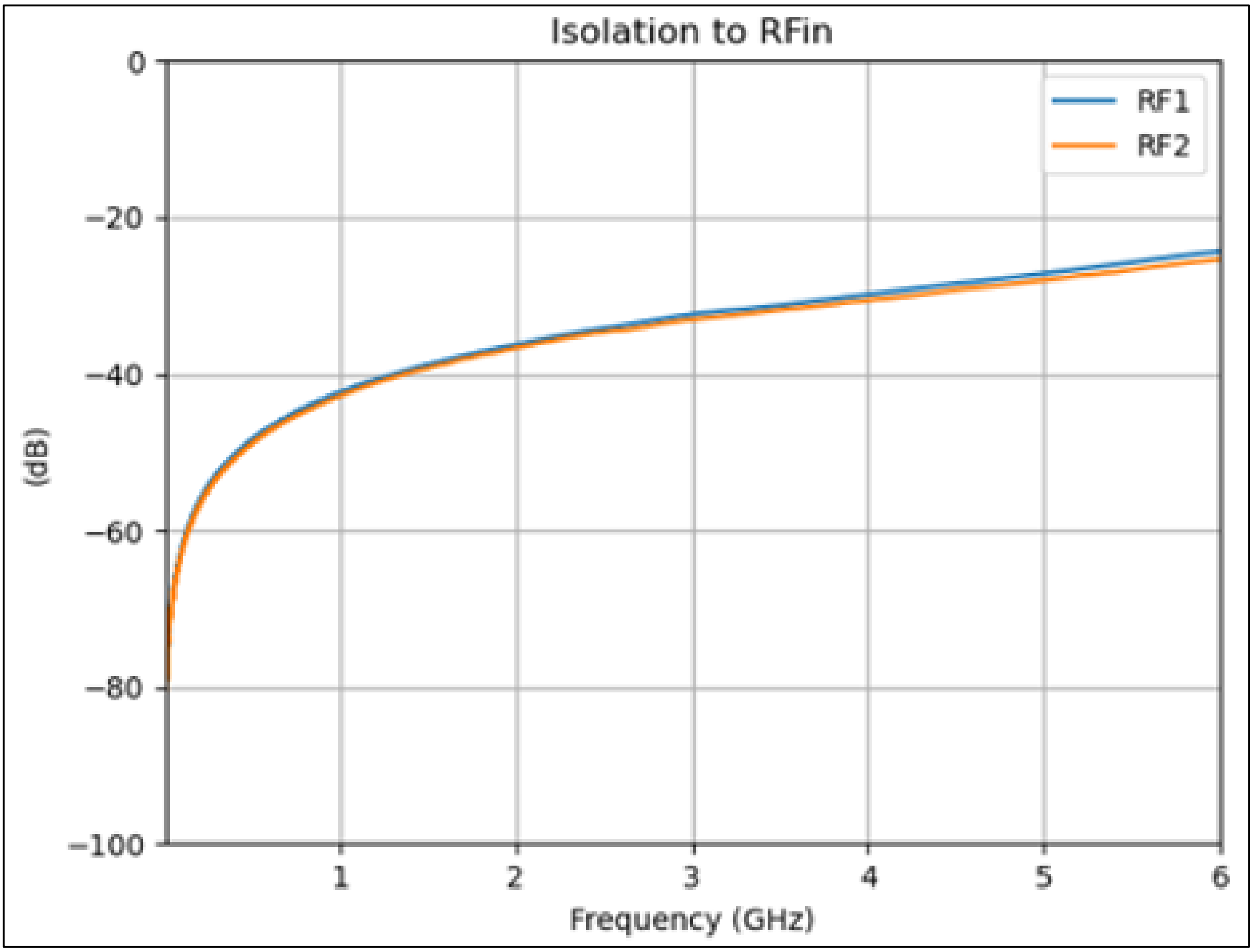


Figure 9 Isolation to RF_{in}

	Isolation to RFIN (dB)						
Frequency (GHz)	**0,96**	**2,17**	**2,7**	**3,8**	**4,2**	**5**	**5,925**
RF Path							
RFIN to RF1 port	42,58	35,15	32,97	30,12	29,05	26,87	24,51
RFIN to RF2 port	42,58	35,18	32,97	30,52	29,33	27,22	24,8

Table 4 Isolation to RF_{in}

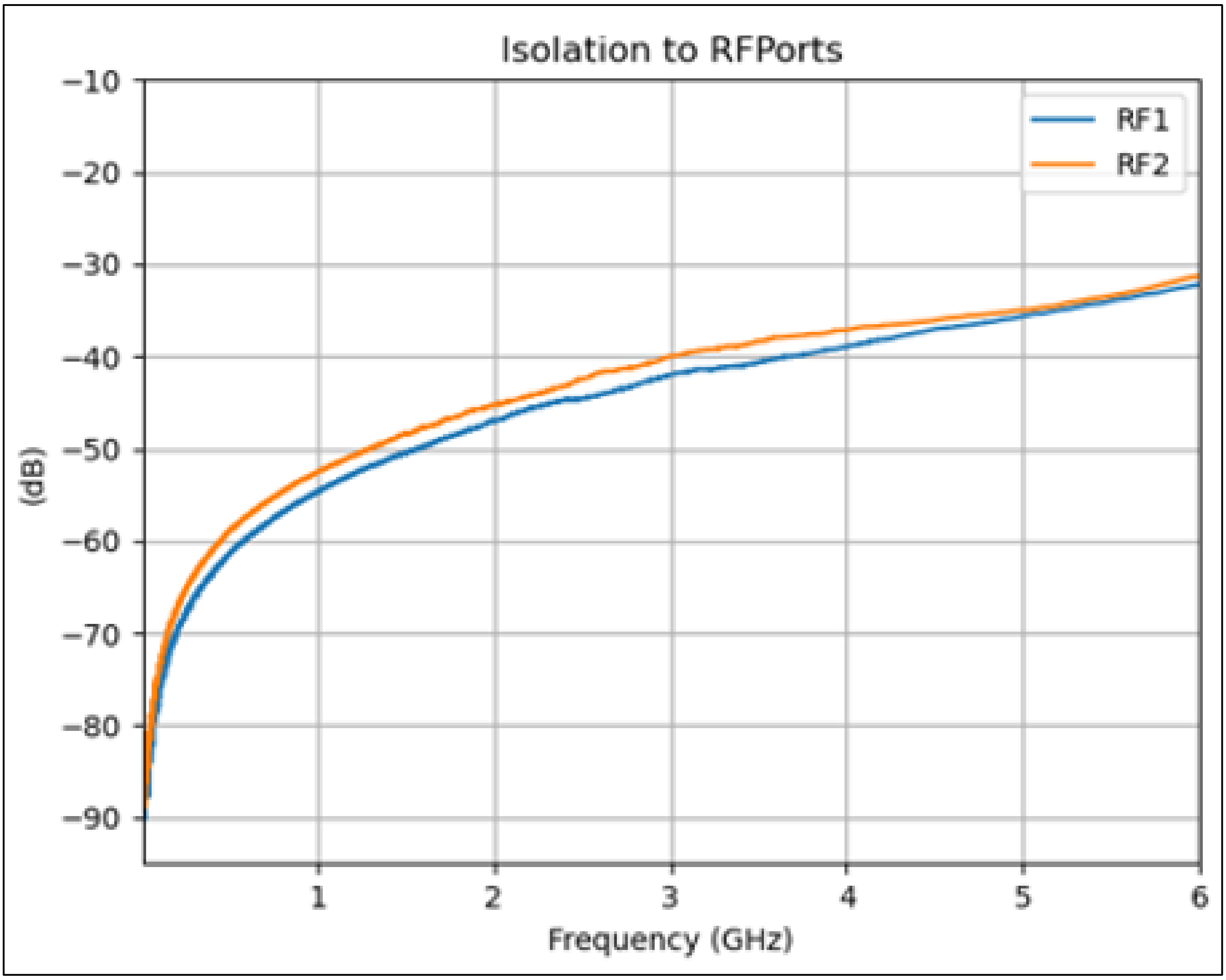


Figure 10 Isolation to RF ports

	Isolation to RF port - port (dB)						
Frequency (GHz)	**0,96**	**2,17**	**2,7**	**3,8**	**4,2**	**5**	**5,925**
RF Path							
RF1	53,75	45,01	42,27	38,53	36,92	33,58	30,46
RF2	54,42	44,47	41,71	37,46	36,64	33,96	30,24

Table 5 Isolation to RF ports

4 De-Embedding

The EVB BGS12P2L6 kit contains a through transmission line that can be used to verify the accuracy of the de-embedding file hold in the webwide within the Eval Board S-Parameter (See image below).

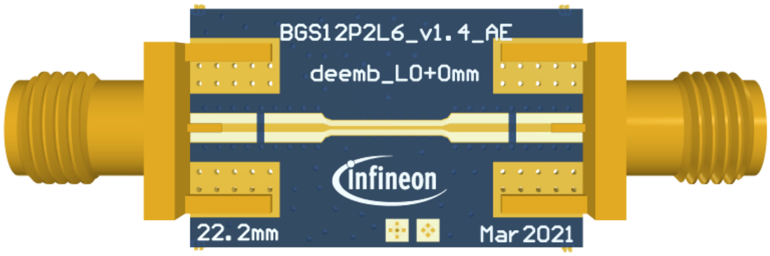


Figure 11 De-Embedding Boards

To generate the De-embedding file, we use the “*2x thru de-embedding*” method. The method is described on the paper “IEEE Standard for Electrical Characterization of Printed Circuit Board and Related Interconnects at Frequencies up to 50 GHz” (IEEE 370-2020).

Abbreviations

EVB	Evaluation Board
SPDT	Single Pole, Double Throw
PCB	Printed Circuit Board
DC	Direct Current
RF	Radio-Frequency
CMOS	Complementary Metal-Oxide Semiconductor
GaAs	Gallium arsenide
ESD	Electrostatic discharge
GPIO	General Purpose Input-Output
SMA	Subminiature Connector
VNA	Vector Network Analyzer

Links

Standard S-Parameters, Datasheet and odder documents can be found here.

Revision history

Document version	Date of release	Description of changes
V1.1	2022-06-03	Final User Guide

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